BIG DATA MANAGEMENT

QuakeWatch

Multi-Country Spatial and Temporal Analysis of Global Earthquakes, 2000–2025

An analysis of 181,600 USGS ComCat records using MongoDB, Python, React and a BI dashboard

A project report submitted for the course
Big Data Management (Semester VII)

Submitted by

Anurag	Roll No.: 2023IMG-008
Kushagra	Roll No.: 2023IMG-030
Kshitij	Roll No.: 2023IMG-029
Vibhor	Roll No.: 2023IMG-054

IMG (Integrated B.Tech + MBA)

ATAL BIHARI VAJPAYEE
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY AND MANAGEMENT GWALIOR
Gwalior, India

November 2026

Source code: `https://github.com/vibhor-5/quakewatch-bdm`

Abstract

QuakeWatch examines when and where earthquakes are recorded around the world over 26 years, and how the share of strong events (magnitude 5.0 or above) varies with country, depth, magnitude type and era. The project uses the U.S. Geological Survey (USGS) ComCat catalogue of earthquakes with magnitude 4.5 or above for 2000–2025, the range over which the global catalogue is usably complete. Python prepares 181,600 tectonic earthquakes, attributes them to 154 countries from the catalogue's place text, and loads three MongoDB collections holding 193,561 documents. A Flask API supplies a React web application in which any year range and any set of countries can be analysed *combined*, and a separate export pipeline prepares seven tables for a business-intelligence dashboard.

Of the 181,600 events, 46,463 are strong, a strong share of 25.59%. Energy is dominated by a handful of great earthquakes: the four largest events (M9.1 Sumatra 2004, M9.1 Tohoku 2011, M8.8 Maule 2010 and M8.8 Kamchatka 2025) radiated 48.0% of all energy released in 26 years, and the largest event of a year carried more than 80% of that year's energy in four years. Indonesia, Japan and Papua New Guinea contribute the most events, and the ten most active countries together account for 59.3% of the catalogue with a pooled strong share of 24.70%. A country's strong share is a one-number summary of the slope of its magnitude distribution (the Gutenberg–Richter b-value): across the 23 countries with at least 1,000 events the correlation between strong share and b is -0.99. Strong share fell from 30.6% in 2006–11 to 22.8% in 2012–18; the number of M6+ events per year fell (178 to 138) while the number of M4.5–4.9 events rose (5,034 to 5,648 per year), and the magnitude-type mix changed, so the catalogue alone cannot separate seismicity from processing. Daily counts are strongly clustered (variance 10.7 times the mean), aftershock counts decay in the Omori pattern, and the yearly b-value rises from about 1.04 to 1.23. A five-run local benchmark records indexed queries examining 38 to 181,600 times fewer documents than collection scans and running in 0–6 ms against 42–142 ms.

Keywords: USGS ComCat, earthquakes, MongoDB, multi-country analysis, Gutenberg–Richter, BI dashboard.

Contents

1 Introduction **3**
- 1.1 Questions addressed . . . 3
- 1.2 Scope . . . 3

2 Data and Preparation **3**
- 2.1 Source and coverage . . . 3
- 2.2 Preparation pipeline . . . 4
- 2.3 Record structure . . . 5

3 Analytical Method **5**
- 3.1 Units and denominators . . . 5
- 3.2 Combining several countries . . . 5
- 3.3 Strong share and the b-value . . . 5
- 3.4 Support and interpretation . . . 5

4 Results and Analysis **6**
- 4.1 Annual totals and magnitude distribution . . . 6
- 4.2 Great earthquakes dominate energy and counts . . . 7
- 4.3 Change in strong share across eras . . . 8
- 4.4 Patterns in the series . . . 9

4.5 Countries . . . 10
4.6 Several countries combined . . . 11
4.7 Depth class, magnitude type, network and review status . . . 12
4.8 Time of day and weekday . . . 13
4.9 Spatial grouping . . . 14

5 Database Design and Query Performance **14**
5.1 Benchmark method . . . 14
5.2 What the plans show . . . 14

6 Web Application **15**
6.1 Global filter bar . . . 15
6.2 Overview and countries . . . 15
6.3 Geographic exploration . . . 17
6.4 Scenario comparison and pattern analysis . . . 18
6.5 Record search and inspection . . . 20

7 BI Dashboard **21**
7.1 Exported tables and their grain . . . 21
7.2 Overview, countries and place, conditions . . . 22

8 Validation and Limitations **24**
8.1 Checks performed for this report . . . 24
8.2 Analytical boundaries . . . 25

9 Recommendations and Conclusion **25**
9.1 Use of the findings . . . 25
9.2 Engineering priorities . . . 25
9.3 Conclusion . . . 26

1 Introduction

Earthquake catalogues combine a physical process (stress release along faults) with a measurement process (seismometer networks, magnitude scales, review procedures) that has itself changed over time. Analysing them demands care about what a number actually counts. The purpose of QuakeWatch is to turn a public catalogue into a usable analytical system: country-level summaries that can be *combined* across any set of countries, long-run time patterns, geographic exploration, condition-based comparison of strong-event share, and inspection of individual events.

1.1 Questions addressed

1. How do event volume, radiated energy and large-event counts vary from year to year over 26 years, and which events dominate?
2. Which countries record the most earthquakes, and how does the strong share differ between countries?
3. What changes when several countries are analysed *combined* rather than one at a time?
4. How do depth class, magnitude type and era change the share of strong events, and which of these differences are physical?
5. How much work do MongoDB indexes save for representative queries, including a geospatial one?
6. How can the same prepared data support both a web application and a BI dashboard?

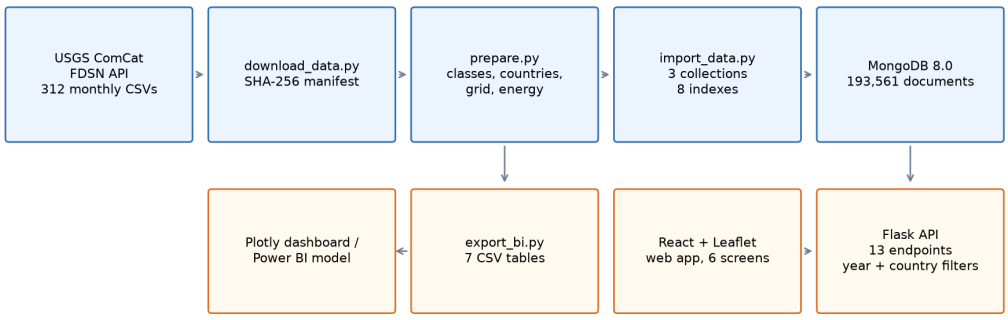


Figure 1: Implemented data flow. The web application reads MongoDB through Flask; the dashboard consumes CSV tables exported from the same prepared data. This is an architecture diagram, not a screenshot.

1.2 Scope

The project demonstrates ingestion, indexing, aggregation and presentation on one local MongoDB instance. It does not demonstrate distributed processing, a production deployment, seismic hazard modelling or earthquake prediction.

2 Data and Preparation

2.1 Source and coverage

The source is the USGS Earthquake Hazards Program ComCat catalogue, accessed through its FDSN event web service.[1] We requested all events with magnitude 4.5 or above for each

calendar month from January 2000 to December 2025 as CSV (312 requests). The cut-off was chosen for completeness: the global network is complete to roughly magnitude 4.5, whereas smaller events are reported fully only in a few countries with dense regional networks, which would make country comparisons reflect network density. The service caps a response at 20,000 rows; the busiest month (March 2011) has 2,401 events, and the downloader aborts if any month approaches the cap.

Table 1: Imported records, checked against MongoDB and the saved import report.

Collection	Content	Documents
`earthquakes`	one document per earthquake	181,600
`daily_summary`	one document per date (2000–2025)	9,497
`country_summary`	one document per country and year	2,464
Total		193,561

The raw download holds 181,683 rows (312 monthly files, no duplicate event identifiers). Eighty-two rows whose event type is not "earthquake" (71 volcanic eruptions, 5 nuclear explosions, 2 explosions, 2 landslides, 1 rock burst and 1 mine collapse) were excluded so that the analysis concerns tectonic events, and one earthquake whose magnitude had been revised below 4.5 after the query window was fixed was dropped. No coordinates were invalid and no magnitude or depth was missing among the 181,600 retained events. The raw CSVs are re-created by the download script and verified against the SHA-256 hashes in the saved manifest; they are not stored in the code repository.

2.2 Preparation pipeline

1. **Download and provenance.** `download_data.py` fetches the months in parallel with retries, saves each CSV and writes a manifest with the request URL, timestamp, size, row count and SHA-256 hash.
2. **Time fields.** Timestamps are converted to UTC; year, month, hour, weekday and date are derived for aggregation.
3. **Classes.** Depth is binned as Shallow (< 70 km), Intermediate (70–300 km) and Deep (> 300 km). Magnitude is binned as Light (4.5–4.9), Moderate (5.0–5.9), Strong (6.0–6.9) and Major (7.0+). An event is *strong* if $M \geq 5.0$; this threshold is unchanged from the project's earlier two-year edition, but because the catalogue now starts at 4.5 the strong share is a fraction of a much larger denominator.
4. **Countries.** The country is the text after the last comma of the USGS `place` string, normalised against a list of 195 country names. US state names are merged into "United States"; a few territories and sub-regions are folded into their country (for example the Kuril Islands into Russia, the Kermadec Islands into New Zealand, Guam and the Northern Mariana Islands into the United States). Events on mid-ocean ridges or in open ocean, whose place text is a region name such as "northern Mid-Atlantic Ridge", are labelled "Ocean / Unassigned". This yields 154 distinct countries; 23,616 events (13.0%) have no country. It is a text heuristic, not a polygon lookup, so an offshore event inherits the country of the nearest named town.
5. **Grid identifier.** Coordinates are floored into $2^\circ \times 2^\circ$ cells and the cell centre forms a grouping key. There are 3,524 non-empty cells. Valid points are stored as GeoJSON (longitude, latitude) for the 2dsphere index.

6. **Energy.** Radiated energy is estimated with the Gutenberg–Richter relation $\log_{10} E = 1.5M + 4.8$ (joules).[4]

2.3 Record structure

Each event is identified by the USGS event id, which is used as the MongoDB `_id`. The two summary collections are derived from it and are rebuilt on every import.

3 Analytical Method

3.1 Units and denominators

For a subset X let n_X be all catalogued events and s_X those with $M \geq 5.0$. Define

$$p_X = \frac{s_X}{n_X}, \qquad \text{strong share}(X) = 100\, p_X. \tag{1}$$

The full 2000–2025 baseline is $p_B = 46{,}463/181{,}600 = 0.2559$. To compare a subset with this baseline we use a share ratio and an odds ratio:

$$R_X = \frac{p_X}{p_B}, \qquad \mathrm{OR}_X = \frac{p_X/(1-p_X)}{p_B/(1-p_B)}. \tag{2}$$

These are ratios of proportions among catalogued events. They do not measure seismic hazard, and because the baseline contains the subset, the comparison is descriptive.

3.2 Combining several countries

When a user selects countries $c_1, \ldots, c_k$ the application pools their events and computes

$$p_{\text{pooled}} = \frac{\sum_i s_{c_i}}{\sum_i n_{c_i}}, \tag{3}$$

which weights each country by its number of events. It is *not* the simple average $\frac{1}{k}\sum_i p_{c_i}$, which would let a country with 1,500 events count as much as one with 22,000. The two differ visibly in Section 4.6. Every API endpoint accepts the same year range and country list, so the pooled view applies to totals, yearly series, maps, scenario comparisons and the record explorer alike.

3.3 Strong share and the b-value

Magnitudes above a completeness threshold M_c follow $N(\geq M) \propto 10^{-bM}$.[4] The maximum-likelihood estimate[5], with the half-bin correction for magnitudes reported to 0.1, is

$$\hat{b} = \frac{\log_{10} e}{\bar{M} - (M_c - 0.05)}, \qquad M_c = 4.5. \tag{4}$$

If the distribution is exactly Gutenberg–Richter, the probability that an event of $M \geq 4.5$ also has $M \geq 5.0$ is $10^{-b\,(5.0-4.5)} = 10^{-b/2}$. The strong share is therefore a one-number summary of the slope of the magnitude distribution. Because $\hat{b}$ is computed from the same events as the share, agreement between the two is largely built in; it shows that the share carries the same information as b, not that b explains the share independently.

3.4 Support and interpretation

Ranked combinations retain groups of at least 100 events; the application's map default is 60 events per cell. No confidence intervals, completeness corrections, causal estimates or out-of-sample predictions are claimed.

4 Results and Analysis

4.1 Annual totals and magnitude distribution

Table 2: Annual event counts, strong share, energy and the weight of the largest event. Energy is in exajoules (EJ, 10^{18} J); the last column is the share of that year's energy radiated by its single largest event.

Year	Events	Strong ($M \geq 5$)	Share	$M \geq 6$	Largest	Energy (EJ)	Top event
2000	5,010	1,494	29.82%	160	M8.0	0.321	19.7%
2001	4,680	1,349	28.82%	137	M8.4	0.436	57.6%
2002	4,963	1,309	26.38%	139	M7.9	0.207	21.6%
2003	5,135	1,366	26.60%	154	M8.2	0.279	39.3%
2004	6,182	1,670	27.01%	157	M9.1	3.016	93.4%
2005	7,878	1,844	23.41%	150	M8.6	0.647	77.5%
2006	7,075	1,879	26.56%	153	M8.3	0.370	48.1%
2007	7,274	2,286	31.43%	196	M8.4	0.724	34.7%
2008	6,307	1,965	31.16%	179	M7.9	0.169	26.4%
2009	6,053	2,072	34.23%	160	M8.1	0.335	26.6%
2010	7,214	2,397	33.23%	174	M8.8	1.229	81.4%
2011	9,584	2,701	28.18%	207	M9.1	3.026	93.1%
2012	7,188	1,698	23.62%	133	M8.6	0.802	62.5%
2013	7,209	1,610	22.33%	142	M8.3	0.412	43.1%
2014	8,333	1,749	20.99%	155	M8.2	0.305	41.3%
2015	7,153	1,571	21.96%	146	M8.3	0.390	45.6%
2016	7,438	1,706	22.94%	147	M7.9	0.293	15.2%
2017	6,371	1,566	24.58%	110	M8.2	0.252	49.9%
2018	7,509	1,768	23.55%	134	M8.2	0.341	37.0%
2019	7,211	1,645	22.81%	145	M8.0	0.160	39.5%
2020	6,510	1,450	22.27%	121	M7.8	0.142	22.3%
2021	8,959	2,227	24.86%	157	M8.2	0.454	27.7%
2022	7,765	1,725	22.22%	127	M7.6	0.108	14.7%
2023	7,643	1,780	23.29%	147	M7.8	0.200	15.8%
2024	6,397	1,507	23.56%	99	M7.5	0.083	13.6%
2025	8,559	2,129	24.87%	145	M8.8	1.209	82.7%
2000–2025	181,600	46,463	25.59%	3,874	M9.1	15.91	–

The catalogue averages 6,985 events per year, from 4,680 (2001) to 9,584 (2011). It holds 3,874 events of magnitude 6 or above (99 to 207 per year, mean 149), 387 of magnitude 7 or above (7 to 24 per year) and 28 of magnitude 8 or above. The magnitude classes contain 135,137 Light, 42,589 Moderate, 3,487 Strong and 387 Major events. The cumulative count falls close to ten-fold per magnitude unit (Figure 3); the maximum-likelihood estimate for the whole catalogue is $\hat{b} = 1.15$, in the usual range for global catalogues, although mixed magnitude types make it only indicative.

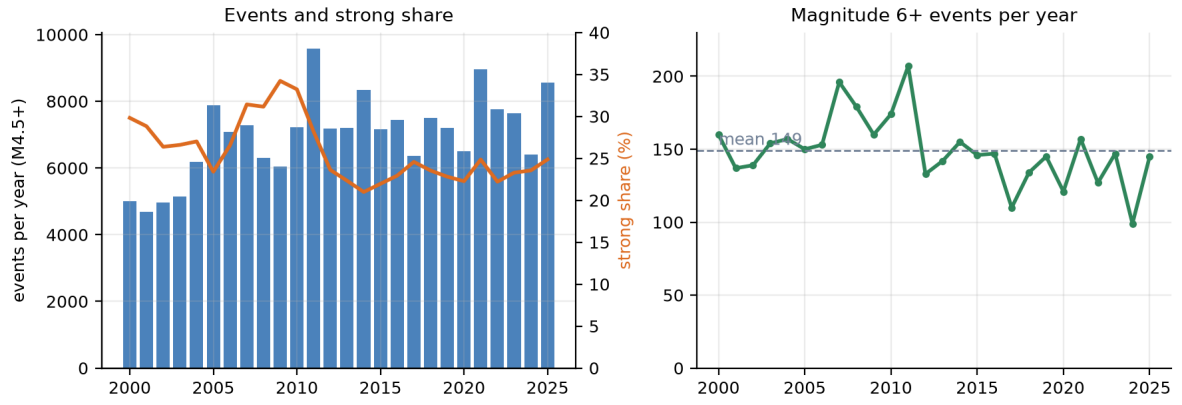


Figure 2: Left: events per year (bars) and strong share (line). Right: M6+ events per year with the 26-year mean. Counts spike after great earthquakes; the strong share moves much more than the M6+ count (Section 4.3).

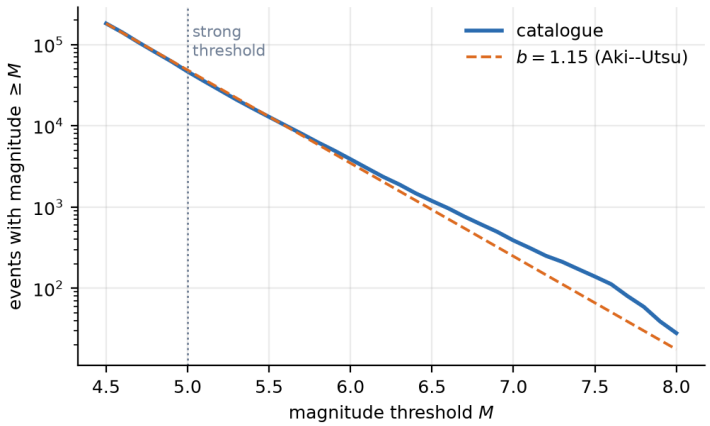


Figure 3: Cumulative number of events at or above each magnitude threshold (log scale) with the fitted Gutenberg–Richter line. The dotted line marks the $M5.0$ strong threshold.

4.2 Great earthquakes dominate energy and counts

Radiated energy is far more concentrated than event counts. Summed over 26 years the catalogue radiates about 15.9 EJ. The four largest events carry 48.0% of it and the ten largest 59.7% (Table 3); in 2004, 2011, 2025 and 2010 the largest event alone contributed 93.4%, 93.1%, 82.7% and 81.4% of that year's energy. The year 2025 radiated 1.21 EJ against 0.083 EJ in 2024, a factor of 14.6, almost entirely because of the M8.8 Kamchatka earthquake of 29 July 2025, even though the event count rose only from 6,397 to 8,559.

Table 3: The six largest events and their share of all energy radiated in 2000–2025.

Date	Event	Magnitude	Share of 26-year energy
26 Dec 2004	Sumatra–Andaman	M9.1	17.7%
11 Mar 2011	Tohoku, Japan	M9.1	17.7%
27 Feb 2010	Maule, Chile	M8.8	6.3%
29 Jul 2025	Kamchatka, Russia	M8.8	6.3%
28 Mar 2005	Off Sumatra (Singkil)	M8.6	3.2%
11 Apr 2012	Wharton Basin	M8.6	3.2%

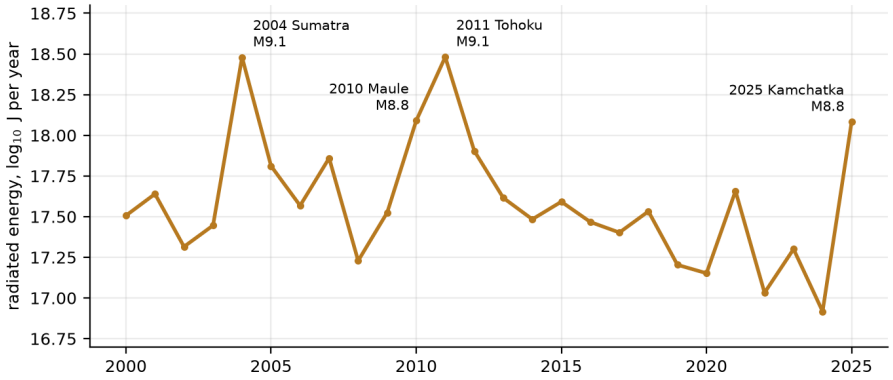


Figure 4: Radiated energy per year (log scale). The four years with the highest energy coincide with the four M8.8–9.1 earthquakes.

Counts react to the same events through aftershocks. The median month holds 552 events; March 2011 holds 2,401 (4.3 times the median) after the Tohoku earthquake, followed by July 2025 (1,321, Kamchatka), August 2021 (1,307), March 2021 (1,262) and January 2005 (1,231, the Sumatra aftershock zone; Figure 5). At country level the effect is larger still: Japan recorded 3,751 events in 2011 against a median year of 539 (7.0 times), Russia 1,905 in 2025 against 211 (9.0 times), Chile 1,271 in 2010 against 206 (6.2 times) and Indonesia 2,668 in 2005 against 750 (3.6 times).

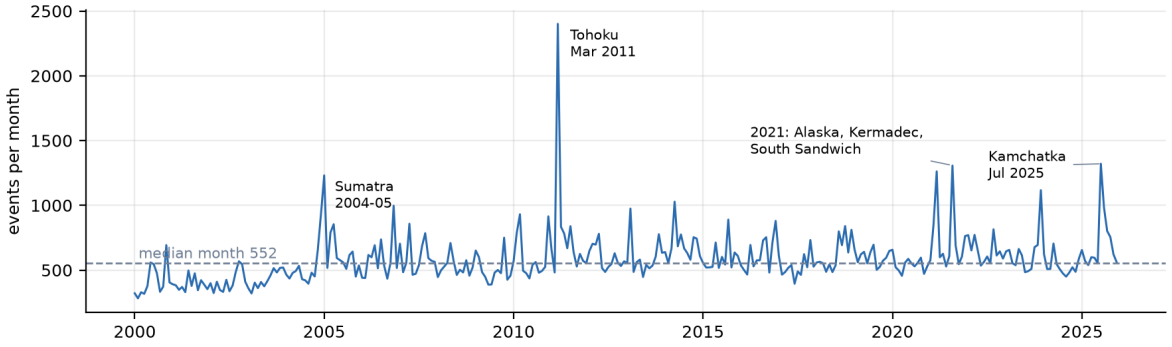


Figure 5: Monthly earthquake counts, 2000–2025. Spikes are aftershock sequences of great earthquakes; the dashed line is the median month.

4.3 Change in strong share across eras

The strong share is not constant over the 26 years (Figure 6, Table 4). It rises from 26.7% in 2000–05 to 30.6% in 2006–11, then falls to 22.8% in 2012–18 and 23.5% in 2019–25. Since share = strong / (strong + light), the table splits the change into its parts. Strong events per year rose from 1,505 to 2,217 and fell to 1,667; M6+ events per year went 150, 178, 138, 134. Light events (M4.5–4.9) per year rose steadily from 4,136 to 5,034, 5,648 and 5,797. So *both* numerators and denominators moved: 2006–11 contains several great-earthquake sequences (Sumatra and Peru and the Solomon Islands in 2007, Maule in 2010, Tohoku in 2011) that inflate the strong count, while the count of M4.5–4.9 events has grown by 40% since 2000.

Table 4: Events per year by era and the resulting strong share.

Era	Events	Per year	M4.5–4.9 per year	$M \geq 5$ per year	$M \geq 6$ per year	Strong share
2000–05	33,848	5,641	4,136	1,505	150	26.7%
2006–11	43,507	7,251	5,034	2,217	178	30.6%
2012–18	51,201	7,314	5,648	1,667	138	22.8%
2019–25	53,044	7,578	5,797	1,780	134	23.5%

The magnitude-type mix changes at the same time (Table 5). The centroid moment type `mwc` supplies 20–23% of events until 2011 and almost none afterwards, while `mww` grows from 0% to 16.5% and `mb` rises to 80–86%. Since magnitude types are not interchangeable at M4.5–5.5, part of the shift in strong share may come from processing rather than from seismicity. We cannot separate the two with the catalogue alone, so cross-era comparisons of strong share should be read with caution and the M6+ count, which is complete in every year, is the safer long-run series.

Table 5: Share of events by magnitude type and era.

Era	`mb`	`mwc`	`mww`	`mwb`	Other types
2000–05	73.9%	20.2%	0.0%	2.8%	3.1%
2006–11	70.4%	23.0%	0.8%	2.1%	3.7%
2012–18	86.3%	0.7%	7.8%	1.4%	3.7%
2019–25	80.3%	0.0%	16.5%	0.1%	3.1%

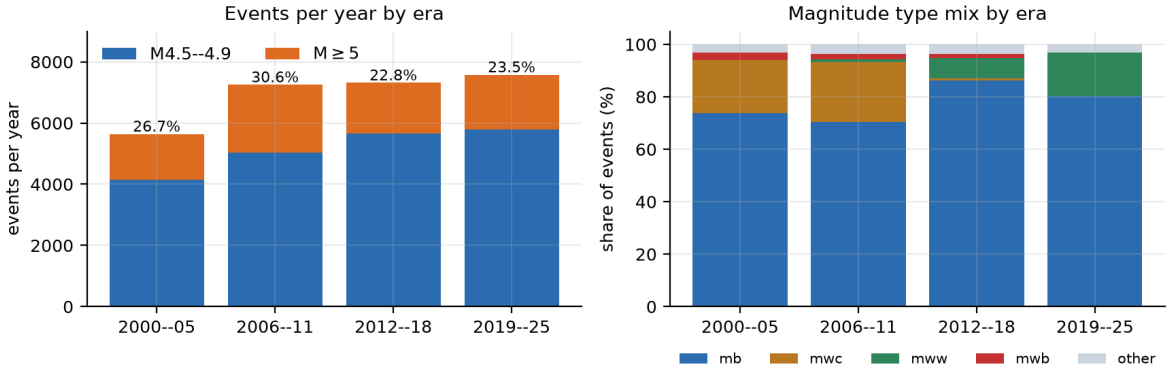


Figure 6: Left: events per year by era, split into M4.5–4.9 and M≥5; labels give the strong share. Right: magnitude-type mix by era.

4.4 Patterns in the series

Three patterns summarise how the catalogue behaves (Figure 7).

Clustering, not randomness. If earthquakes arrived independently at the average rate of 19.1 per day, the variance of the daily count would equal its mean. The observed variance is 10.7 times the mean, and the distribution has a heavy right tail (the last bar pools days with 39 or more events). The busiest day, 11 March 2011, has 454 events. The busiest 1% of days hold 6.1% of all events and the busiest 10% hold 23.2%, so activity arrives in bursts that follow large earthquakes.

Aftershock decay. After the Tohoku earthquake the Japan-attributed catalogue records 531 events in the first day and 13 on day 30; after the Kamchatka earthquake the Russia-attributed catalogue records 425 and 3. On log–log axes the daily counts fall along a straight line, the signature of the Omori law, with fitted exponents of about 1.07 (Tohoku) and 1.28 (Kamchatka)

over the first 30 days. These exponents are indicative only: they come from a simple regression on country-attributed counts without declustering.

A rising b-value. The yearly b-value is lowest in 2009 (0.99) and highest in 2014 (1.29); its mean is 1.04 over 2000–04 and 1.23 over 2021–25. A higher b means proportionally more small events, which is exactly the shift behind the falling strong share in Section 4.3. Whether this reflects the catalogue (more M4.5–4.9 events detected or re-labelled) or the Earth cannot be decided from these data.

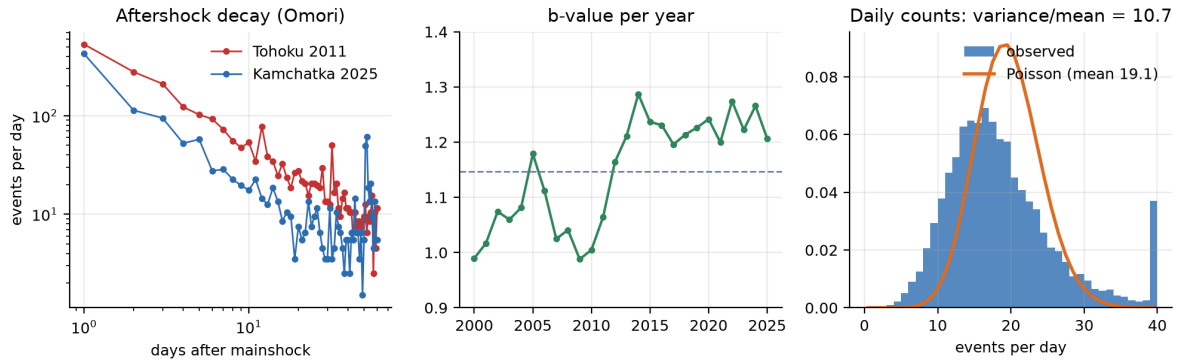


Figure 7: Left: events per day after the Tohoku and Kamchatka mainshocks (log–log). Centre: yearly b-value with the 26-year value dashed. Right: observed daily counts against a Poisson distribution with the same mean.

4.5 Countries

The prepared catalogue assigns 157,984 events to 154 countries; 60 countries have at least 100 events and 23 at least 1,000. Table 6 lists the 15 most active. Indonesia leads with 22,230 events (12.2% of the catalogue), followed by Japan (16,228), Papua New Guinea (10,581), New Zealand (9,672) and the United States (9,367, including Alaska and Hawaii). The ten most active countries together hold 107,673 events (59.3%) and a pooled strong share of 24.70%, close to the world's 25.59%, so the headline figure is not distorted by one country.

Table 6: The 15 most active countries, 2000–2025. b is the Aki–Utsu estimate; the last column is mean depth in km.

Country	Events	Strong	Strong share	b	Largest	Depth
Indonesia	22,230	5,619	25.28%	1.16	M8.6	55
Japan	16,228	3,527	21.73%	1.23	M9.1	48
Papua New Guinea	10,581	2,961	27.98%	1.08	M7.9	59
New Zealand	9,672	2,373	24.53%	1.19	M7.8	48
United States	9,367	2,509	26.79%	1.12	M7.9	51
Philippines	8,910	1,925	21.60%	1.27	M7.6	65
Russia	8,407	2,032	24.17%	1.18	M7.8	48
Tonga	8,226	2,176	26.45%	1.13	M7.6	75
Chile	7,116	2,122	29.82%	1.05	M8.3	45
Fiji	6,936	1,350	19.46%	1.30	M7.9	434
Vanuatu	5,686	1,678	29.51%	1.03	M7.8	82
Solomon Islands	4,354	1,334	30.64%	1.03	M7.8	39
China	2,880	646	22.43%	1.26	M7.9	19
New Caledonia	2,834	875	30.88%	1.01	M7.7	20
Peru	2,800	729	26.04%	1.10	M8.4	69

Among the 23 countries with at least 1,000 events the strong share ranges from 19.5% (Fiji) to 31.4% (Mexico). Mexico, New Caledonia (30.9%), the Solomon Islands (30.6%), Chile

(29.8%) and Vanuatu (29.5%) are at the top; Fiji, Timor Leste (19.5%), Argentina (19.7%), Greece (20.6%) and the Philippines (21.6%) at the bottom. The relationship with b is almost exact (Figure 8): the correlation between strong share and $\hat{b}$ is -0.99, and no country lies more than 2.8 percentage points from $100 \cdot 10^{-b/2}$. As explained in Section 3, this is largely by construction, so the useful reading is that a country's strong share is a compact descriptor of its magnitude slope. Depth adds a partial physical hint: the three lowest-share countries Fiji, Argentina and Timor Leste have mean depths of 434, 154 and 157 km, and across the 23 countries the correlation between share and mean depth is -0.48, consistent with deep events having a lower strong share (20.4%) than shallow ones (26.9%; Section 4.7). We do not claim that depth causes the difference.

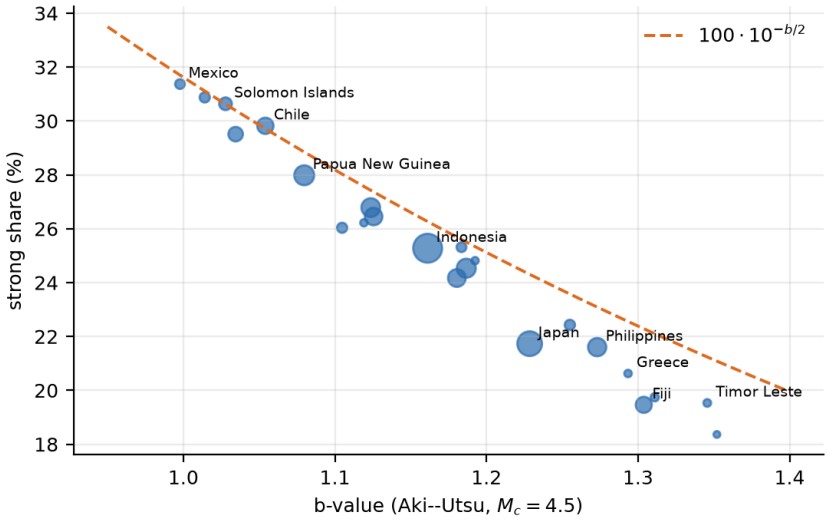


Figure 8: Strong share against b-value for the 23 countries with at least 1,000 events (marker area proportional to event count). The dashed curve is the Gutenberg–Richter relation $100 \cdot 10^{-b/2}$.

The 23,616 events without a country (open ocean and ridges) have a strong share of 29.2%, higher than any large country, and are excluded from country tables but included in world totals.

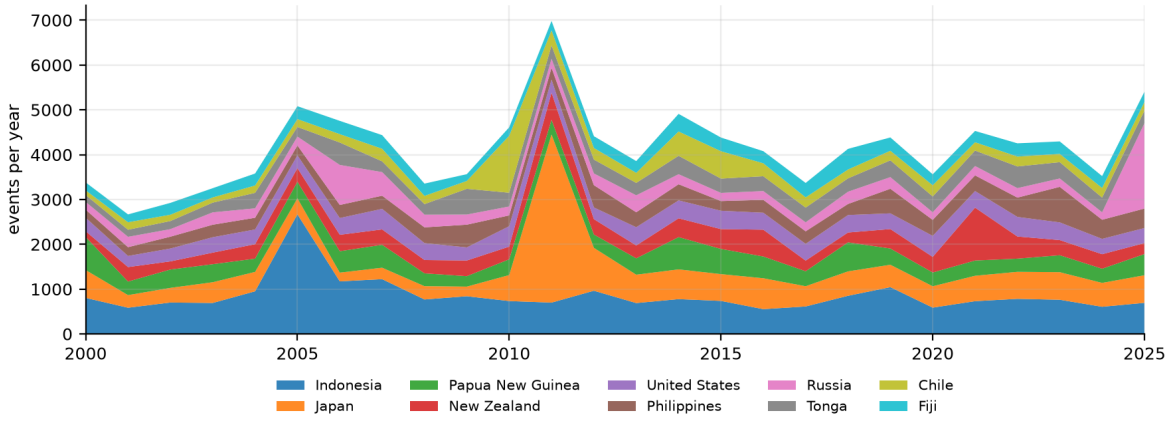


Figure 9: Events per year for the ten most active countries, stacked. The tallest peak is 2011 (Japan); the 2005 Indonesian peak and the 2025 Russian peak are also visible.

4.6 Several countries combined

The application ships four preset groups, and any other set can be built by clicking country rows. Table 7 compares them with the world baseline.

Table 7: Preset country groups, pooled across all events. "Average" is the unweighted mean of the countries' strong shares; b is for the pooled events. R and OR are against the 25.59% world baseline.

Group	Countries	Events	Strong	Pooled share	Average	b	R	OR
Ring of Fire (core)	10	97,874	24,601	25.14%	25.93%	1.16	0.98	0.98
South Pacific islands	5	28,036	7,413	26.44%	27.39%	1.11	1.03	1.05
Alpide belt	11	12,993	3,006	23.14%	23.67%	1.23	0.90	0.88
Latin America	12	19,640	5,315	27.06%	25.30%	1.10	1.06	1.08

Ring of Fire (core: Japan, Indonesia, Philippines, Papua New Guinea, Chile, Peru, Mexico, New Zealand, the United States and Russia) holds 97,874 events, 53.9% of the catalogue, at 25.14%. The South Pacific islands (Tonga, Fiji, Vanuatu, Solomon Islands, New Caledonia) are slightly above the baseline at 26.44%, Latin America at 27.06%, and the Alpide belt (Turkey to Myanmar, including Iran, India, China and Greece) is lowest at 23.14% with the highest b (1.23). Pooling matters: for Latin America the pooled share (27.06%) is 1.8 points above the unweighted average (25.30%) because Mexico, with 31.4% strong share, is large, while for the Ring of Fire it is 0.8 points below the average because Japan and the Philippines, with low shares, carry many events (Figure 10).

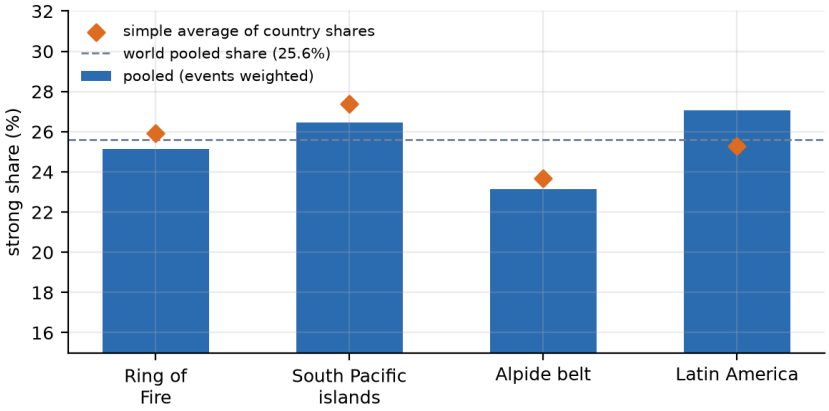


Figure 10: Pooled strong share of the four preset groups (bars) against the unweighted average of their countries' shares (diamonds). The dashed line is the world share.

4.7 Depth class, magnitude type, network and review status

Table 8: Strong share by depth class.

Depth class	Events	Strong	Share
Shallow (< 70 km)	143,549	38,656	26.93%
Intermediate (70–300 km)	29,607	6,083	20.55%
Deep (> 300 km)	8,444	1,724	20.42%

Shallow events have the highest strong share (26.93%) and intermediate and deep events are nearly equal at about 20.5%. Within a single magnitude type the ordering persists: for `mb`, the most common type, the strong share is 12.43% for shallow (111,375 events), 7.85% for intermediate and 6.97% for deep events, a ratio of 1.78 between shallow and deep.

Strong share depends dramatically on how the magnitude was computed (Table 9, Figure 11).

Table 9: Strong share by magnitude type (types with at least 800 events).

Magnitude type	Events	Strong	Share
`mwb` (moment, body wave)	2,630	2,629	99.96%
`mww` (moment, W-phase)	13,150	11,680	88.82%
`mwc` (moment, centroid)	17,169	14,834	86.40%
`mw` (moment)	844	449	53.20%
`mwr` (moment, regional)	3,415	401	11.74%
`mb` (body wave)	142,406	16,220	11.39%
`ml` (local)	1,127	118	10.47%

These differences are not physical. The USGS computes moment magnitudes (`mwb`, `mww`, `mwc`) mainly for larger events, so almost every event with such a type is strong, whereas `mb` is used for the bulk of M4.5–5.0 events. The `us` network contributes 179,756 events (99.0%) with a strong share of 25.45%, so network comparisons are dominated by one contributor; the 122 events with status *automatic* (0.07%) have a share of 59.8% against 25.56% for the 181,478 reviewed events, a group too small to interpret. Among depth-by-type groups with at least 100 events the strong share ranges from 6.02% (shallow `md`, 249 events) to 100% (deep `mwb`, 130 events).

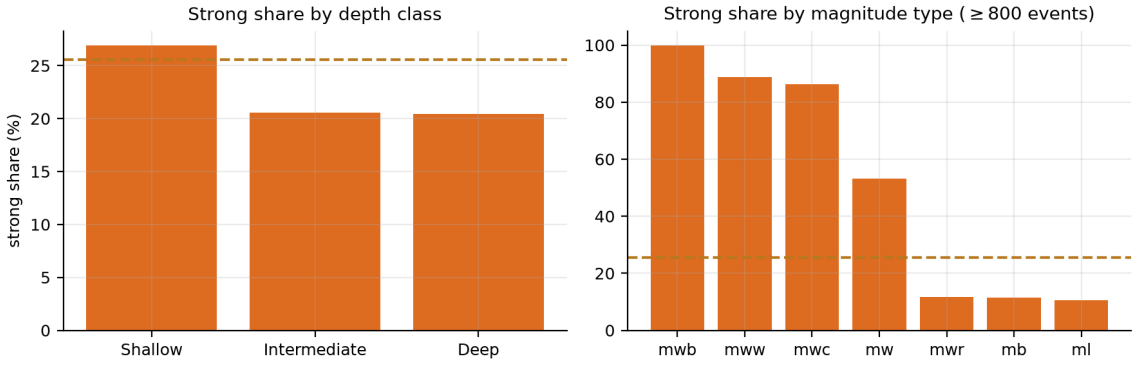


Figure 11: Strong share by depth class and magnitude type. The dashed line marks the 25.59% baseline.

4.8 Time of day and weekday

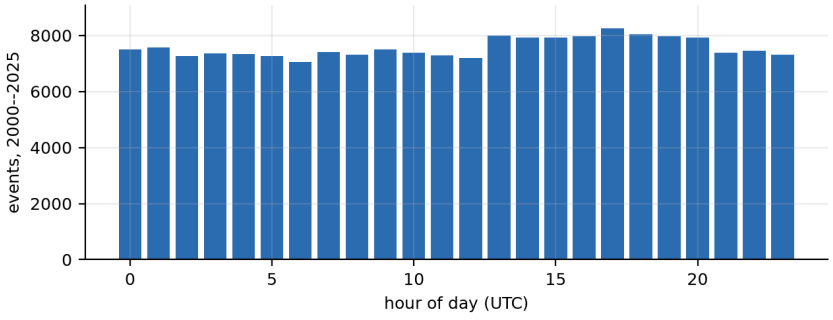


Figure 12: Events by hour of day (UTC), 2000–2025.

Hourly counts range only from 7,054 (06:00 UTC) to 8,252 (17:00 UTC), within about ±8% of the mean of 7,567, and weekday counts from 25,560 to 26,331. Weekday strong share ranges from 24.91% (Tuesday) to 26.56% (Wednesday); with a few dominant sequences each falling on one weekday, this spread should not be read as a weekly cycle. Earthquakes follow no human

schedule, which is a useful contrast with traffic or crime data: the flat profile is the expected result and a sanity check on the pipeline's time handling.

4.9 Spatial grouping

The prepared data has 3,524 non-empty 2° cells; the dashboard map keeps the 907 cells with at least 30 events, and the application's default of 60 events leaves 573. The busiest cell is centred at 5°S, 153°E (the New Britain and Bougainville region of Papua New Guinea) with 2,182 events, of which 593 (27.2%) are strong; the second is centred at 29°S, 177°W (the Kermadec trench) with 1,879. A 2° latitude interval is about 222 km, but longitude spacing shrinks with latitude, so cells are angular bins, not equal-area cells or the output of a clustering algorithm.

5 Database Design and Query Performance

The `earthquakes` collection has the default `_id` index (the USGS event id) plus indexes on year–strong flag, date–strong flag, grid–year, magnitude, country–year, country–strong flag, and a `2dsphere` index on the GeoJSON `location`. The summary collections carry unique indexes on date and on country–year.

5.1 Benchmark method

The benchmark uses MongoDB 8.0.34 in a local container on 181,600 documents. Each query runs five times with a named index hint and five times with a `$natural` (collection scan) hint, using `explain` with `executionStats`. Median server-reported times and the plan stages are saved.[2]

Table 10: Local benchmark: median of five executions per access path.

Query	Access path	Keys	Docs	Returned	ms
2011 strong events	compound index	2,701	2,701	2,701	1
	collection scan	0	181,600	2,701	52
Japan strong events	compound index	3,527	3,527	3,527	1
	collection scan	0	181,600	3,527	53
Event id lookup	`_id` index	1	1	1	0
	collection scan	0	181,600	1	42
Within 300 km of Tokyo	2dsphere index	4,820	4,797	4,197	6
	collection scan	0	181,600	4,197	142
Magnitude ≥ 6	magnitude index	3,874	3,874	3,874	2
	collection scan	0	181,600	3,874	46

5.2 What the plans show

Each indexed plan is an `IXSCAN` below a `FETCH`; each alternative is a `COLLSCAN`. The indexed queries examine 67 times fewer documents (2011 strong events), 52 times fewer (Japan strong events), 47 times fewer (magnitude ≥ 6), 38 times fewer (300 km query) and 181,600 times fewer (identifier lookup). The compound index `country_strong` serves the country filter used by the combined-country views. Reported times of 0–2 ms lie near the timer's whole-millisecond resolution, so no speed-up ratio is computed from them; the scan times of 42–142 ms rank in the order of documents examined and per-document work (the geographic filter is the most expensive to evaluate).

For the geospatial query the index returns 4,820 candidate keys for 4,197 matches, because it covers the bounding cells of the spherical cap; the remaining candidates are filtered after fetch. This contrasts with an index that scans its whole range: a bounded `$geoWithin/$centerSphere`

predicate gives the 2dsphere index real selectivity.[3] The same index serves the record inspector, which looks for events within 150 km and one day of a selected quake.

These timings describe this local run on one dataset size, not a service guarantee or asymptotic scaling. At 181 thousand documents all queries are still fast even without an index, so absolute savings are small; the document-count ratios are the more transferable evidence.

6 Web Application

React and TypeScript provide the interface, Flask provides 13 HTTP endpoints, and MongoDB aggregation pipelines compute summaries. The six screens are selected with buttons; they are not separate URL routes. The captures below were taken from the running application by selecting each screen and waiting for data to load.

6.1 Global filter bar

A filter bar above the tabs holds a year range (2000–2025), an "add country" selector, a preset-group selector and removable chips for the chosen countries. The filter is passed to every endpoint as `year_from`, `year_to` and `countries`, so each screen shows the pooled result for exactly the selected years and countries; with no country chosen the whole world is analysed.

6.2 Overview and countries

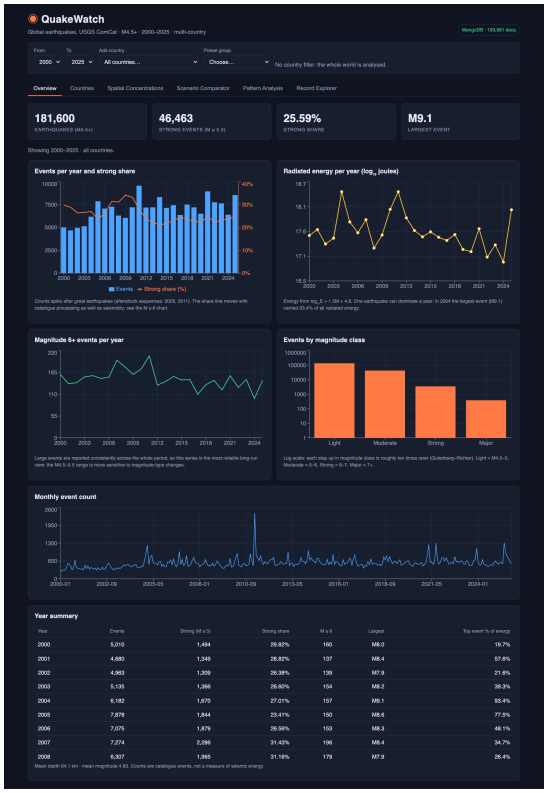


Figure 13: Actual Overview screen, all years and countries: 181,600 events, 46,463 strong events (25.59%) and the M9.1 maximum, with events and strong share per year, energy per year, M6+ events per year, the monthly series and the year summary.

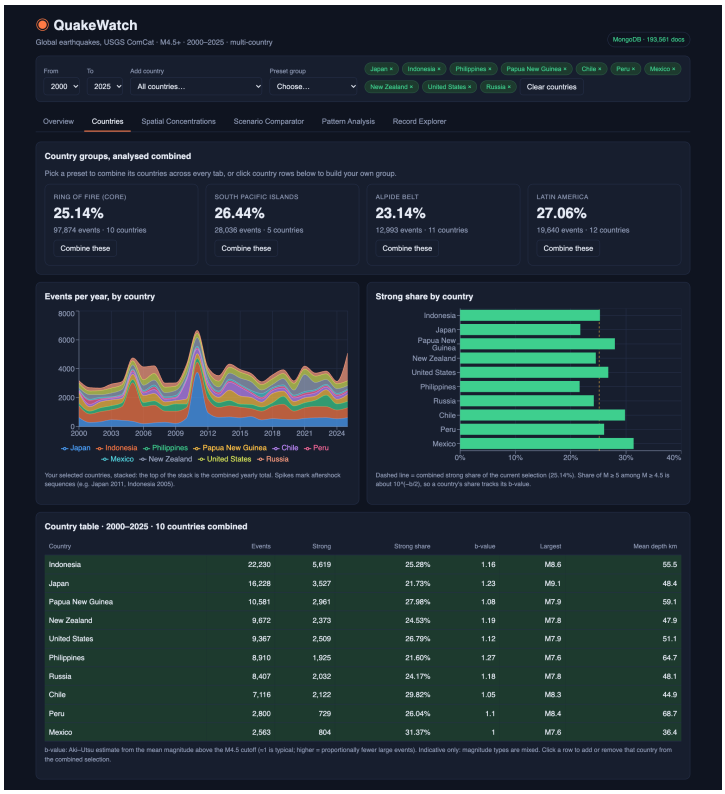


Figure 14: Actual Countries screen with the Ring of Fire (core) preset applied: the ten countries appear as chips in the filter bar, the stacked chart shows their yearly events, the bars show each country's strong share against the pooled dashed line, and the table lists events, strong share, b-value, largest event and mean depth. Clicking a row adds or removes that country.

6.3 Geographic exploration

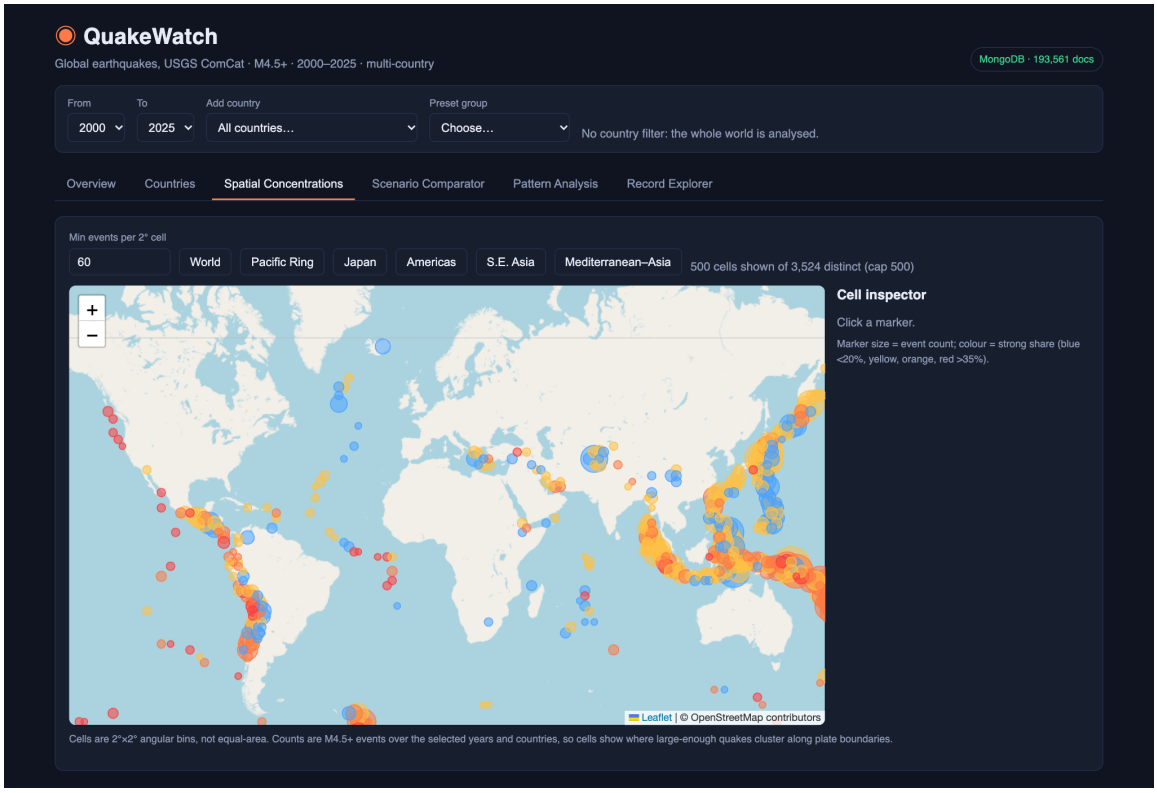


Figure 15: Actual Spatial Concentrations screen, all years, with a 60-event minimum: 500 of 3,524 cells are shown (the request cap). Marker size is event count and colour is strong share.

The map uses Leaflet with OpenStreetMap tiles. Users change the per-cell minimum, jump to a region such as the Pacific Ring, Japan or the Mediterranean–Asia belt, and inspect a cell's counts and mean depth. The request is capped at 500 cells, which is reported on screen. The pattern of cells follows plate boundaries, from the Aleutians and Japan through the Philippines and Indonesia to Tonga and the Kermadec trench, and along the Andes.

6.4 Scenario comparison and pattern analysis

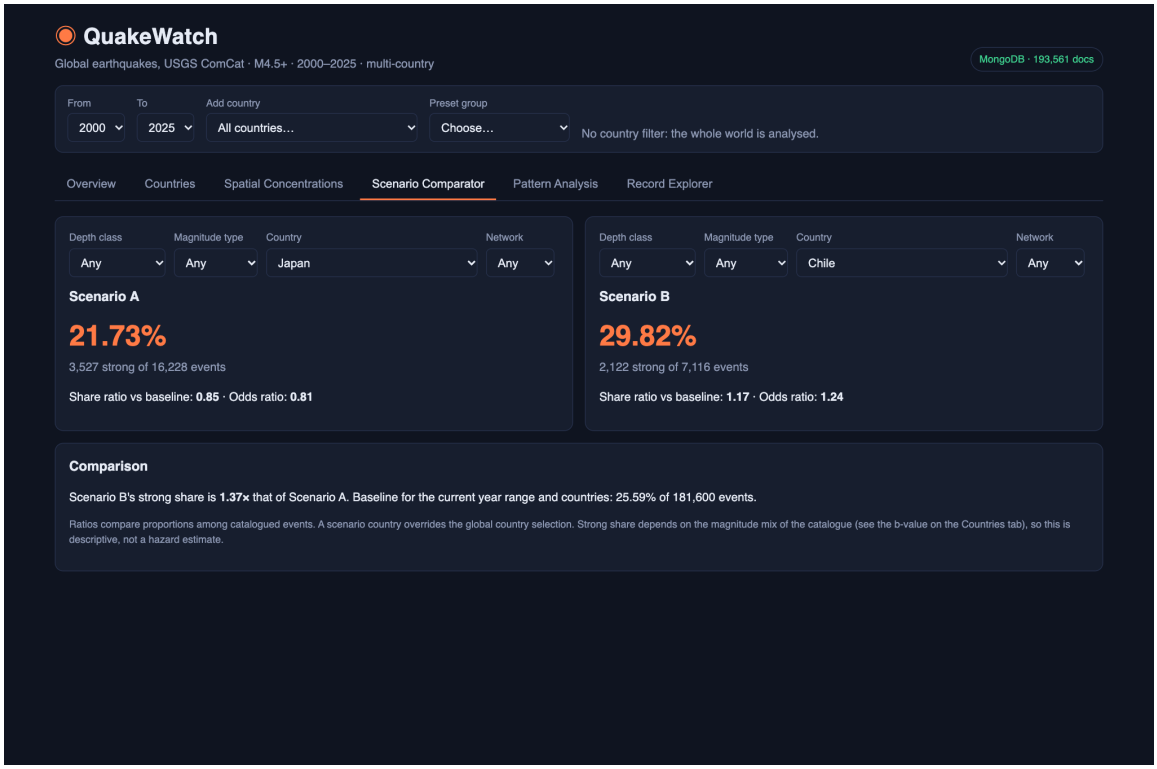


Figure 16: Actual scenario comparator, comparing Japan (21.73%) with Chile (29.82%): Chile's strong share is 1.37 times Japan's.

Users choose country, depth class, magnitude type and network for two scenarios. The API computes each subset's counts and ratios from the same boolean flag and baseline, and the interface shows sample counts next to every percentage. A scenario country overrides the global country selection.



Figure 17: Actual Pattern Analysis screen: strong share by depth class and magnitude type, hourly counts, weekday strong share, depth×type combinations and the most active countries.

6.5 Record search and inspection

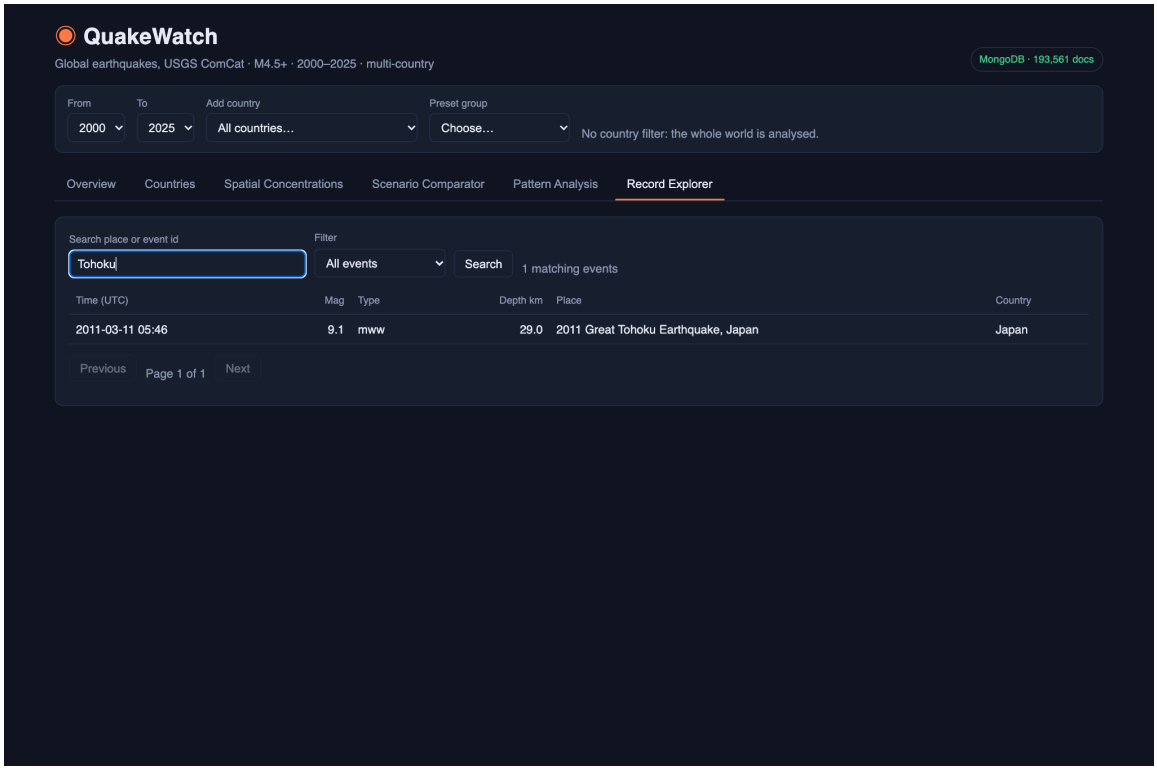


Figure 18: Actual Record Explorer after searching “Tohoku”.

Results are paginated in groups of 25 and can be searched by place or event id. Opening an event retrieves the stored document and runs a geospatial query for events within 150 km and one day, capped at 50 results.

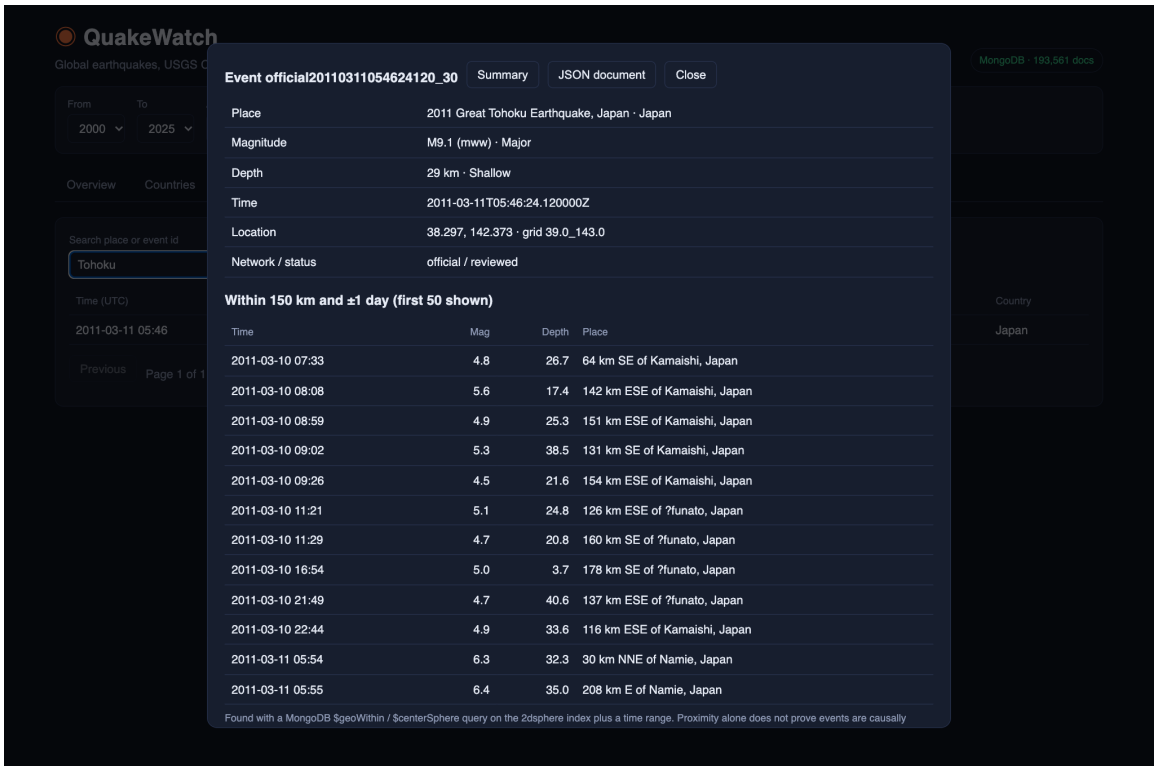


Figure 19: Actual inspector for the 2011 Tohoku earthquake (M9.1, depth 29 km), listing nearby events in the preceding and following day, including the M5.6 and M6.3 events around the mainshock; the list is capped at 50. A second tab shows the stored JSON document.

7 BI Dashboard

7.1 Exported tables and their grain

The export pipeline writes a star schema as CSV files.

Table 11: Dashboard tables verified from the exported CSV files.

Table	Rows	Grain
`fact_earthquakes`	181,600	One event
`dim_calendar`	9,497	One date
`dim_country_group`	38	Preset group and country (bridge)
`agg_hotspots_grid`	8,265	Grid cell and year (≥ 5 events)
`agg_conditions`	672	Condition type, value and year
`agg_countries`	1,437	Country and year (≥ 5 events)
`agg_depth_mag_matrix`	302	Depth class, magnitude class and year

The same tables can be loaded into Microsoft Power BI Desktop. The repository includes the DAX measures (event count, strong events, strong share, baseline share, share and odds ratios, M6+ events, country count, b-value, total energy and a year-aware grid measure using `TREATAS`) and step-by-step build instructions, including a many-to-many bridge so that a slicer on `dim_country_group` combines several countries. The dashboard shown below is the bundled HTML build produced from the same CSV files with Plotly; it mirrors the three-page layout a Power BI report would use.

7.2 Overview, countries and place, conditions

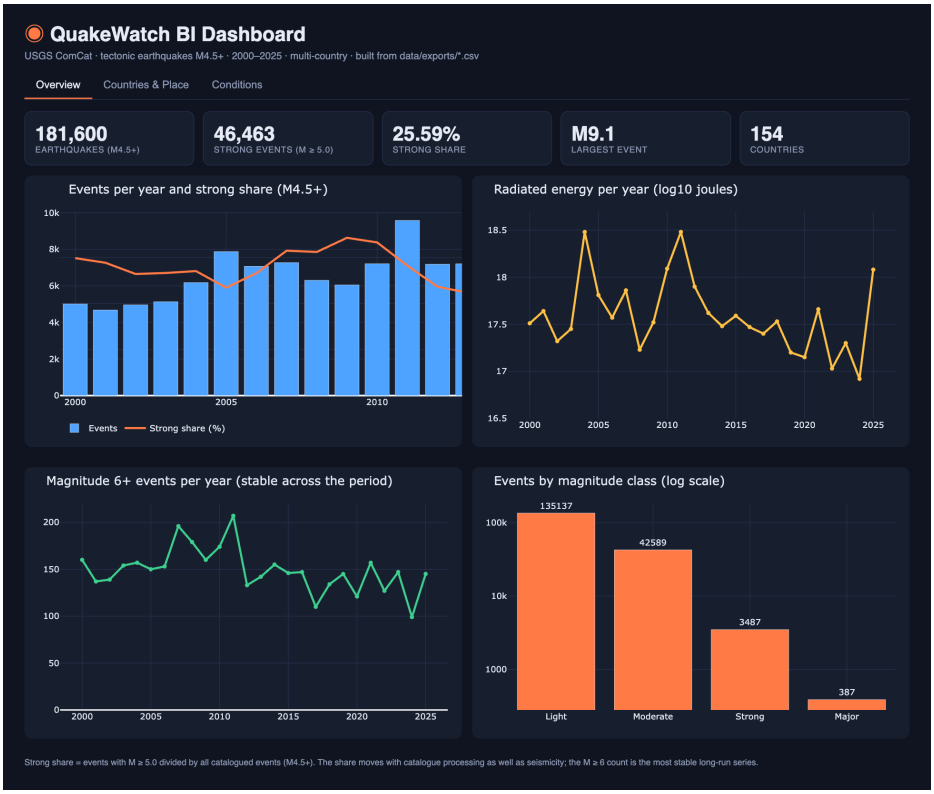


Figure 20: Dashboard overview page: 181,600 events, 46,463 strong events, 25.59% share, M9.1 maximum, 154 countries, events and strong share per year, energy per year, M6+ events per year and a log-scale magnitude-class chart.

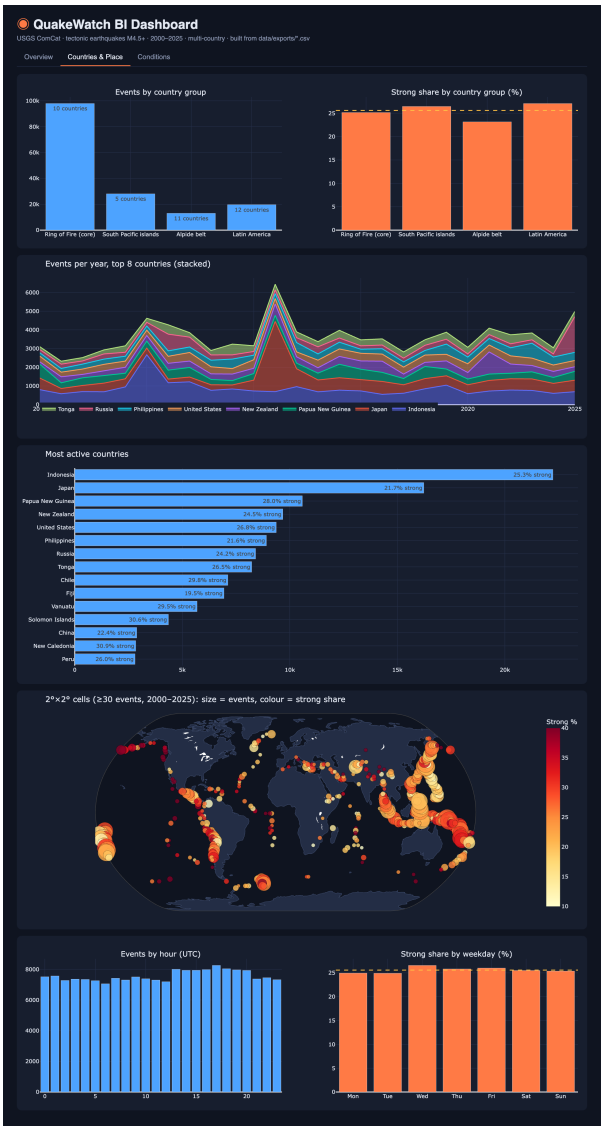


Figure 21: Dashboard countries and place page: events and strong share by country group, stacked events per year for the top eight countries, the most active countries, a world map of grid cells (size is events, colour is strong share), and hourly and weekday charts.

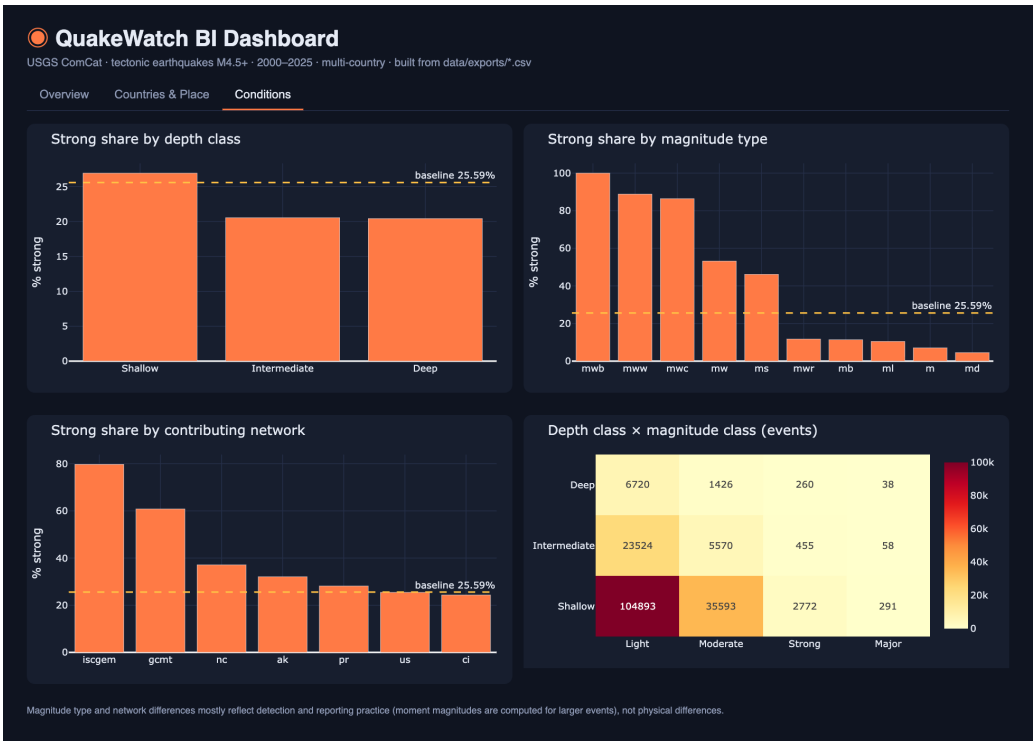


Figure 22: Dashboard conditions page: strong share by depth class, magnitude type and contributing network, and a depth-by-magnitude heat map.

8 Validation and Limitations

8.1 Checks performed for this report

Table 12: Verification status at report preparation, November 2026.

Check	Observed result
Import report	181,683 raw rows; 82 non-earthquakes and 1 sub-M4.5 event excluded; 0 duplicates; 0 invalid coordinates; 181,600 retained; 154 countries; 23,616 events without a country.
Collection counts	181,600 + 9,497 + 2,464 = 193,561 documents, matching the live database.
Severity reconciliation	Strong events 46,463 of 181,600; annual totals sum to the combined totals.
Magnitude classes	135,137 + 42,589 + 3,487 + 387 = 181,600.
Country attribution	Per-country counts plus 23,616 unassigned events sum to 181,600; a test checks that a two-country selection equals the sum of its parts.
API tests	13 passed (country parsing, totals, year-range partition, multi-country sums, *b*-value and strong-share consistency, trends, groups, baseline ratios, temporal sums, combination ratios, pagination, detail and 404).
Frontend build	TypeScript check and Vite build completed; a bundle-size warning remains.
Screenshots	Distinct web and dashboard pages, visually reviewed.
Benchmark evidence	Five timings per query and access path; plan stages retained.

Passing tests establishes only what the tests exercise. Report figures were also recomputed independently in Python (pandas) from the raw files and agree with the API and database.

8.2 Analytical boundaries

- **Catalogue completeness.** Magnitude 4.5 is the practical global completeness limit, but it was not reached equally in all countries and years; early-period counts of M4.5–4.9 events are likely lower than later ones for reasons of detection and processing, which feeds into the strong share.
- **Era effects.** Strong share changes between eras partly through magnitude-type processing (Section 4.3). We report the decomposition but cannot attribute the change to seismicity or to processing.
- **Magnitude type.** Strong share by magnitude type reflects when each type is computed, not physics.
- **Country heuristic.** Countries come from parsing the `place` text. Offshore events inherit the nearest named town's country, 13.0% of events have no country, and the folding of territories (for example Guam into the United States, the Kermadec Islands into New Zealand) is a modelling choice.
- ***b*-value.** The estimate uses one cut-off for all countries, mixed magnitude types and no uncertainty interval, and shares its data with the strong share, so their agreement is not independent evidence.
- **Dominant events.** A few great earthquakes (and their aftershocks) strongly affect yearly counts, shares and energy; 26 years is long enough to see their effect but too short to establish a trend in great-earthquake rates.
- **Angular grid.** 2° cells are not equal-area, and the map shows at most 500 cells per request.
- **Nearby events.** Events within 150 km and one day are candidate aftershocks or foreshocks; no sequence model is fitted and the list is capped at 50.
- **Dashboard format.** The BI dashboard shipped is an HTML build; the Power BI model is described by the exported tables and DAX measures, and a `.pbix` file must be assembled in Power BI Desktop.
- **Statistics.** No confidence intervals, magnitude-of-completeness estimation or hazard modelling are performed.

9 Recommendations and Conclusion

9.1 Use of the findings

Long-run and cross-country comparisons of strong share should be made within one magnitude type and one catalogue contributor, or replaced by counts above a higher completeness threshold (M6+ is complete in every year). When countries are combined, pooled shares should be used, never unweighted averages of percentages.

9.2 Engineering priorities

1. Replace the text-based country heuristic with country and exclusive-economic-zone polygons through MongoDB geospatial joins (`$geoIntersects`).

2. Estimate the magnitude of completeness per country and period and report completeness-adjusted rates with confidence intervals for b.
3. Add equal-area (for example hexagonal) cells.
4. Identify aftershock sequences with a declustering algorithm instead of a fixed window, so that background and sequence rates can be separated.
5. Assemble and publish the Power BI `.pbix` with the provided measures and add a scheduled refresh from the USGS service.

9.3 Conclusion

QuakeWatch demonstrates a path from a public earthquake catalogue to a queryable document database, an interactive web application in which any set of countries and years is analysed combined, and a BI dashboard. The verified data contains 181,600 earthquakes from 2000 to 2025 in 154 countries with a strong share of 25.59%. The analysis separates frequency from magnitude and from energy, shows four great earthquakes radiating 48% of all energy, treats a country's strong share as a summary of its magnitude slope, exposes how reporting practice shapes apparent patterns across magnitude types and eras, and shows why combined views must be pooled. Profiling demonstrates that indexes, including the 2dsphere index with a bounded geographic predicate, cut the documents examined by between one and five orders of magnitude. The limitations above define the boundary within which the findings can be presented.

References

[1] U.S. Geological Survey. *ComCat / FDSN Event Web Service*. Earthquake Hazards Program. Accessed 12 November 2026. `https://earthquake.usgs.gov/fdsnws/event/1/`

[2] MongoDB. *Explain Results*. Execution statistics and query-plan stages. `https://www.mongodb.com/docs/manual/reference/explain-results/`

[3] MongoDB. *2dsphere Indexes*. Geographic query support and GeoJSON indexing. `https://www.mongodb.com/docs/manual/core/indexes/index-types/geospatial/2dsphere/`

[4] B. Gutenberg and C. F. Richter. Magnitude and energy of earthquakes. *Annali di Geofisica*, 9:1–15, 1956.

[5] K. Aki. Maximum likelihood estimate of b in the formula $\log N = a - bM$ and its confidence limits. *Bulletin of the Earthquake Research Institute, University of Tokyo*, 43:237–239, 1965.